The Russo-Ukrainian War: An Analysis of the Russian Air/Drone Strikes on Ukraine

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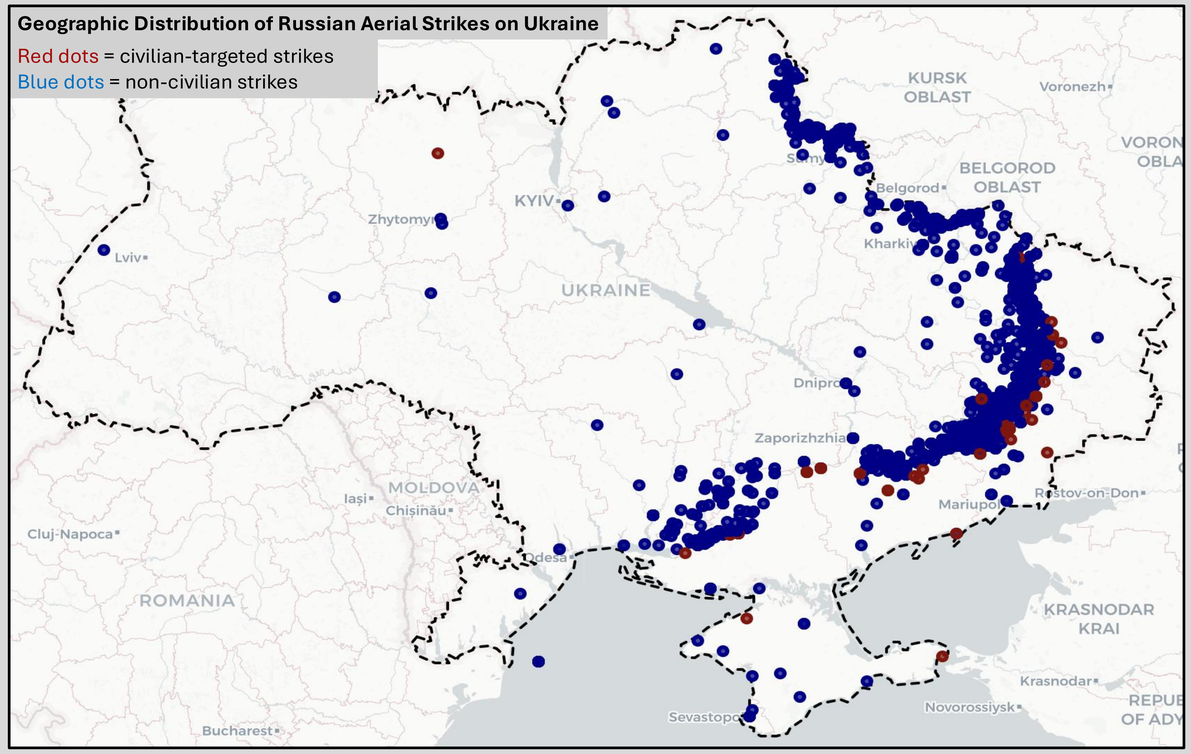
Introduction

Russian air and drone strikes are a defining feature of the Russo-Ukrainian war, causing widespread destruction and civilian casualties. Understanding the scale and impact of these attacks is crucial for policymakers, humanitarian organizations, and conflict analysts. Despite existing datasets, integrated analysis linking Unmanned Aerial Vehicle (UAV) patterns with broader conflict dynamics remains limited. This project’s goal is to understand geographic and fatality patterns based on past airstrike data.

1.7
Fatalities per strike

1,030
Number of locations affected



HESA Shahed 136
Iranian UAV accounting for 88% of airstrikes

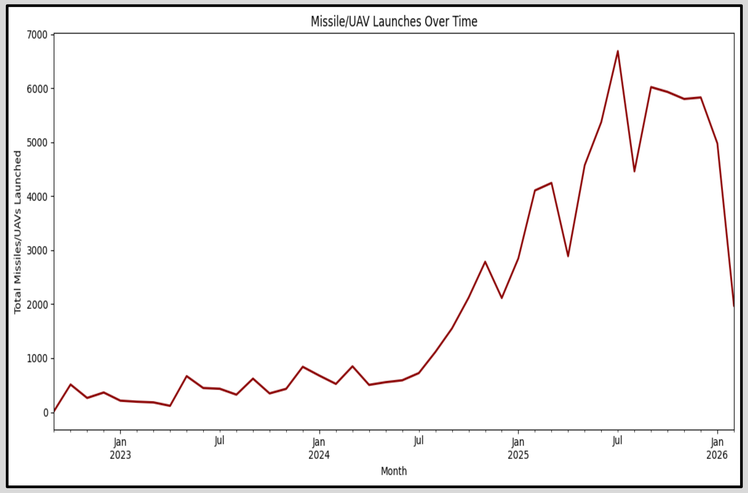
Key Findings & Conclusion

Our investigation into the Ukraine conflict, with a special focus on UAV strikes, has uncovered significant insights. The war involves a substantial number of events, with over 6,000 airstrikes directly leading to over 10,000 deaths. Geographically, certain areas like Donetsk and Luhansk on the eastern region, closer to the battlefront of the war, consistently endure a disproportionately high number of airstrikes, correlating with greater casualties. Our predictive analysis further confirms that **where and when these events occur** are crucial factors in determining their severity and the number of casualties. This predictive capability offers a powerful tool for understanding and potentially mitigating the human cost of these attacks. Understanding the intricacies of the war in Ukraine has many societal and ethical implications. Our hope with this research is to meaningfully contribute to the ongoing analysis of the war effects in Ukraine.

Data Sources (2022-2026)

1. Armed Conflict Location & Event Data (ACLED)
   - Conflict events, actors, locations, and fatalities across Ukraine

2. UAV attack data & model specifications (Kaggle Open Data)
   - Ukrainian developer tracking media outlets and posting data to Kaggle
   - Detailed records of launched/destroyed missiles & UAVs, including model info and target



Donetsk
Most affected region

85,814
Total recorded airstrikes in last 4 years

Check out our interactive 3D map!

Modeling

After multiple iterations and evaluation, “Random Forest” (RF) model selected to predict presence of casualties on given airstrike for increased interpretability. Top features:

- Day of year
- Longitude/Latitude of target
- Population of target city

RF Model Evaluation

Using 80%-20% training and test split, model was cross-validated using standard metrics. Results below, showing successful and accurate prediction of fatality presence for given strike.

Accuracy = 0.90 Recall = 0.90

F1-Score = 0.90 Precision = 0.89

Societal Implications

Humanitarian Aid
- Identifying high-risk areas and understanding casualty patterns can inform targeted humanitarian efforts.

Policy & Defense
- Insights into attack types and locations can aid in developing more effective defense strategies and resource allocation.

Conflict Monitoring
- Predictive models can contribute to early warning systems for escalating violence and influence military and policy responses.